Pre-Analysis Plan – #ManyDaughters

Our identification strategy builds on the exogenous variation in the sex of the first-born child: We estimate the effect of having a first-born daughter instead of a first-born son. Any consequence arising from having a first-born daughter instead of a first-born son is thus understood as part of the causal effect of having a daughter. For example, if having a first-born daughter increases the propensity to get a second child and having a second child influences gender equality attitudes, this causal chain originated from having a daughter instead of a son.

We acknowledge that adoption and modern fertility methods create the opportunity to influence the child's sex, introducing potential endogeneity. Since the numbers of adoptions and children conceived with the help of in vitro fertilization are low in Germany (see, e.g., Statistisches Bundesamt [Destatis], 2025, and von Wolff & Haaf, 2020, respectively) and German law allows sex selection for the latter only under special medical conditions, we do not believe that this aspect poses a significant threat to our identification strategy. An additional shortcoming is that depending on the variable used, a relationship between a child and a parent may be misspecified (e.g., as biological) in the SOEP data, as described in the codebook for the data set biobirth. There are several complementary analyses that would allow evaluating the robustness of the proposed analyses (e.g., for some participants in the SOEP, data on whether the child has been adopted is available). In line with the project requirements, we refrain from specifying additional analyses and focus on a single analysis for each hypothesis here.

The hypotheses formulated by the project coordinators always refer to the effect of having "daughters." We assume that this formulation does not imply that the project coordinators want the research teams to focus on the effect of having *several* daughters vs. no daughter at all. Instead, we interpret this formulation as referring to the effect of having (one or several) daughters. By exploiting the variation in the sex of the first-born child, we focus on the part of the hypotheses that we believe can be causally identified without additional strong assumptions.

Since our research team did not have access to any data when preparing the pre-analysis plan, we do not have a good estimate of the sample sizes for our proposed tests. We used the response statistics for the separate variables reported online to get a rough estimate of how many observations we may expect. Yet, we had no information on how invalid responses are correlated across variables. In an attempt to avoid underpowered statistical inference (particularly in Hypotheses 1 and 4, where only data from 2012 is used), we pre-register not to interpret the statistical test from a specific analysis if its effective sample size falls below 1500. The lack of more precise information on how including certain variables reduces the effective sample size also encouraged us to refrain from incorporating a large set of control variables.

Hypothesis 1

Outcome variable: We create an index based on the following variables from pl that were elicited (only) in 2012: plh0301 (level of agreement with statement that women should rather care about family than career), plh0303 (level of agreement with statement that men should be equally involved in housework), and plh0307 (level of agreement with statement that men should care more about financial security of family than females). We employ the numeric values of the answer categories (from 1: agree completely to 4: completely disagree) and reversely code the second variable such that higher values always reflect more support for gender equality. We then standardize each variable by subtracting the mean and dividing by the standard deviation of this variable in the group of households with a first-born son. Afterward, we calculate the average of the standardized variables on the individual level, which constitutes our main outcome variable.

Analysis: We conduct an OLS regression, where we regress the outcome variable on a dummy for whether the first-born child is female (based on kidsex01 in biobirth). We additionally include a dummy for whether the respondent is male (based on pla0009_h in pl) and a categorical variable for the age of the respondent in 2012 (based on ple0010_h in pl, calculated as difference between 2012 and the year of birth and assigned to categories: 18–24/25–30/31–40/ 41–50/51–64/65 or older) as regressors. We cluster standard errors on the household level (based on hid in pl). To test the hypothesis, we test whether we can reject that the coefficient of the dummy for the first-born child being female is smaller than or equal to zero at the 5% significance level (based on a one-sided t-test).

Exclusion criteria: We exclude individuals who do not have children or are below 18 years old based on our categorical age variable. We additionally exclude individuals whose first-born child was born in 2012 – the year the outcome variable has been elicited – or later (i.e., individuals are excluded if kidgeb01 in biobirth is not strictly smaller than 2012). In general, we exclude individuals who have a missing value (i.e., no valid response) for any of the variables used.

Hypothesis 2

Outcome variable: Average value of variable plh0004 (own assessment of being left- or right-leaning on a 10-point scale) on the individual level, considering all observations in the years after the first child was born (based on kidgeb01 in biobirth) and treating the answer categories as numeric.

Analysis: We conduct an OLS regression, where we regress the outcome variable on a dummy for whether the first-born child is female (based on kidsex01 in biobirth). We additionally include a dummy for whether the respondent is male (based on pla0009_h in pl), a categorical variable for the age of the respondent in the survey year of the first considered observation of plh0004 (based on syear and ple0010_h in pl, calculated as difference to year of birth and assigned to categories: 18–24/25–30/31–40/ 41–50/51–64/65 or older), and a dummy for the year of the first considered observation of plh0004 (based on syear) as regressors. We refrain from considering values of variable plh0004 before the birth of the first child as these may not be available for many respondents and thus would substantially reduce sample size. We cluster standard errors on the household level (based on cid in pl). To test the hypothesis, we test whether we can reject that the coefficient of the dummy for the first-born child being female is greater than or equal to zero at the 5% significance level (based on a one-sided t-test).

Exclusion criteria: We exclude individuals who do not have children or are below 18 years old based on our categorical age variable. In general, we exclude individuals who have a missing value (i.e., no valid response) for any of the variables used.

Hypothesis 3

Outcome variable: We use the allocation of working hours within households where parents of a first-born child live together (as a couple). In particular, we identify all individuals with a first-born son or daughter (based on kidsex01 in biobirth) and their household members (based on hid in pl) in the first year the individual is observed in the data after the child was born (based on syear in pl and kidgeb01 in biobirth). If a household identified by this procedure includes a couple (based on pid and pgpartnr in pgen) that is of different sex (based on pla0009_h in pl) and shares the same first-born child (based on kidpnr01 or, if not available, kidsex01 and kidgeb01 in biobirth), we calculate the following outcome variable for this household: Across the years after the year of

birth of the first-born child where the parents are still together in a household (based on hid in pl), none of the parents is a pensioner (based on pgstib in pgen), data on working hours (see variables used below) is available for both, and no other couple is part of the household (based on hid, pid, and pgpartnr), we calculate the average actual working hours (based on pgtatzeit in pgen, set to zero for individuals with a missing value who are not employed based on pgstib in pgen) per parent. Afterward, we divide the average working hours of the female parent by that of the male parent (identifying parent sex based on pla0009_h in pl). The outcome variable on the household level is defined as the absolute deviation of this ratio from 1.

Analysis: We conduct an OLS regression, where we regress the outcome variable on a dummy for whether the first-born child is female (based on kidsex01 in biobirth). We additionally include a separate categorical variable for the age of each parent in the survey year of the first considered observation of their working hours (based on ple0010_h in pl, calculated as difference to year of birth and assigned to categories: 18–24/25–30/31–40/ 41–50/51–64/65 or older), and a dummy for the year of the first considered observation of their working hours (based on syear) as regressors. We use heteroskedasticity robust standard errors. To test the hypothesis, we test whether we can reject that the coefficient of the dummy for the first-born child being female is greater than or equal to zero at the 5% significance level (based on a one-sided t-test).

Exclusion criteria: The construction of the outcome variable implies that we only focus on households that include a different-sex couple with the same first-born child. Households that include more than one couple are excluded (including other household members in addition to one couple does not lead to exclusion, but only the couples' working hours will be considered). The outcome variable also implies that we only consider the working hours for the period of the first observation after the year of birth of the first-born child until the couple is no longer together in a household, one of them becomes a pensioner, for one of them the working hours are no longer observed, or another couple enters the household. In addition, we exclude households where at least one of the parents is below 18 years old based on our categorical age variable.

Hypothesis 4

Outcome variable: See Hypothesis 1.

Analysis: We conduct the same regression as for Hypothesis 1 but add the interaction of the dummy for the first-born child being female and the dummy for the respondent being male. To test the hypothesis, we test whether we can reject that the coefficient of the interaction term is smaller than or equal to zero at the 5% significance level (based on a one-sided t-test). We thus focus on the part of Hypothesis 4 that the project coordinators presented in the inequality expression. The verbal description of Hypothesis 4 may instead require that the interaction term is positive AND that for both, males and females, having a daughter leads to an increase.

Exclusion criteria: See Hypothesis 1.

References

Statistisches Bundesamt [Destatis] (2025). Adoptierte Kinder und Jugendliche: Deutschland, Jahre, Geschlecht, Nationalität. Retrieved on June 26, 2025, from https://www-genesis.destatis.de/datenbank/online/statistic/22521/table/22521-0001

von Wolff, M., & Haaf, T. (2020). In Vitro Fertilization Technology and Child Health. *Deutsches Ärzteblatt International*, 117(3), 23–30. https://doi.org/10.3238/arztebl.2020.0023